Exploratory Data Analysis (EDA) Report

Project Title: Retail Sales Analysis

Introduction

In this project, I performed **Exploratory Data Analysis (EDA)** on the Retail Sales dataset. The main purpose of this analysis is to understand the sales data, clean the dataset, and identify important patterns. EDA helps in understanding the data before building a machine learning model or making business decisions.

Objective

The objectives of this project are:

- To understand the Retail Sales dataset.
- To check the dataset structure and data types.
- To identify missing values and duplicate records.
- To calculate basic statistical measures.
- To detect outliers in sales data.
- To visualize the sales data using different charts.

Dataset Description

First, I imported the required libraries such as **Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn**. Then I loaded the Retail Sales dataset using Pandas. I used functions like `head()`, `tail()`, `shape`, `columns`, `info()`, and `describe()` to understand the dataset. These functions helped me know the number of rows, columns, data types, and summary statistics.

Data Cleaning

After loading the dataset, I checked for missing values using `isnull().sum()`. I also checked for duplicate records using `duplicated().sum()` and removed duplicate rows using `drop_duplicates()`. Outliers in the **Sales (venda)** column were detected using the **Interquartile Range (IQR)** method. This helped improve the quality of the dataset.

Statistical Analysis

I calculated the following statistical measures for the **Price (preco)** column:

- Mean
- Median
- Mode
- Variance
- Standard Deviation

I also calculated the **correlation matrix** to understand the relationship between numerical columns.

Data Analysis

I performed some filtering and sorting operations on the dataset:

- Displayed records where **Sales (venda) > 100**.
- Displayed products where **Price (preco) > 2**.
- Sorted the dataset based on the **Stock (estoque)** column.

These operations helped me understand the sales performance and product details.

Data Visualization

To understand the dataset better, I created the following graphs:

- **Bar Chart** – Shows sales for different dates.
- **Line Chart** – Displays the sales trend over time.
- **Histogram** – Shows the distribution of sales values.
- **Scatter Plot** – Shows the relationship between price and sales.

These graphs made it easy to identify trends and patterns in the sales data.

Observations

From the analysis, I observed the following:

- The dataset was loaded successfully.

- Missing values and duplicate records were checked.
- Outliers in the sales column were identified using the IQR method.
- Statistical measures provided a summary of the price column.
- The sales trend changed over different dates.
- The scatter plot showed the relationship between price and sales.
- The visualizations helped in understanding overall sales performance.

Conclusion

This project helped me understand how to perform Exploratory Data Analysis on a Retail Sales dataset. I learned how to clean the data, calculate statistical measures, detect outliers, perform filtering and sorting, and create different visualizations. Through this analysis, I gained a better understanding of sales trends and product performance. After completing the EDA process, the dataset is ready for further analysis and machine learning model development.